NATIONAL UNIVERSITY OF PAKISTAN
Department of Computer Science

DIGITAL LOGIC DESIGN

PROJECT REPORT

Railway Crossing Flasher

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Course:	Digital Logic Design
Platform:	Arduino Uno (Tinkercad Simulation)

1. Abstract

The Railway Crossing Flasher is a Digital Logic Design project that simulates the warning signal system used at railway level crossings. The system uses an Arduino Uno microcontroller to alternately flash a red LED and a yellow LED, replicating the visual warning pattern seen on real railway crossing signs when a train is approaching. The project combines digital logic concepts, embedded programming, circuit simulation, and 3D physical design to produce a complete, working model of a real-world safety system.

2. Introduction

Railway level crossings use alternately flashing red lights mounted on a signal post to warn road users that a train is approaching and that it is unsafe to cross the tracks. This project recreates that warning mechanism using basic digital electronics: two LEDs (red and yellow) controlled by a microcontroller so that only one LED is ever lit at a time, alternating at a fixed interval.

The project was implemented and simulated using Tinkercad Circuits, with an Arduino Uno driving two LEDs through current-limiting resistors on a breadboard. A physical enclosure resembling a real railway crossing signal post was also designed in Tinkercad's 3D design environment, giving the project both a functional electronic component and a realistic physical form.

3. Objectives

- To design a digital circuit that alternately flashes two LEDs at a fixed time interval.
- To program an Arduino microcontroller to control digital output pins using timed delays.
- To simulate and verify the circuit behavior using Tinkercad Circuits before physical implementation.
- To design a realistic 3D enclosure representing a railway crossing signal post.
- To apply digital logic and embedded systems concepts to a real-world safety application.

4. Components Used

S.No	Component	Quantity	Purpose
1	Arduino Uno	1	Microcontroller to control LED timing
2	Red LED	1	Represents one crossing warning light
3	Yellow LED	1	Represents the second crossing warning light
4	Resistors (220Ω)	2	Current limiting for LEDs
5	Breadboard	1	Circuit assembly platform

S.No	Component	Quantity	Purpose
6	Jumper Wires	As required	Connections between Arduino and breadboard
7	USB Cable	1	Power and code upload

5. Pin Configuration

Arduino Pin	Connected To
Pin 4	Red LED (anode, via 220Ω resistor)
Pin 3	Yellow LED (anode, via 220Ω resistor)
GND	Breadboard negative rail (LED cathodes)

6. Circuit Diagram

The circuit was designed and simulated in Tinkercad Circuits. The Arduino Uno's digital pins 4 and 3 are connected to the red and yellow LEDs respectively, each through a 220Ω current-limiting resistor, with the LED cathodes returning to the Arduino's GND pin through the breadboard's negative rail.

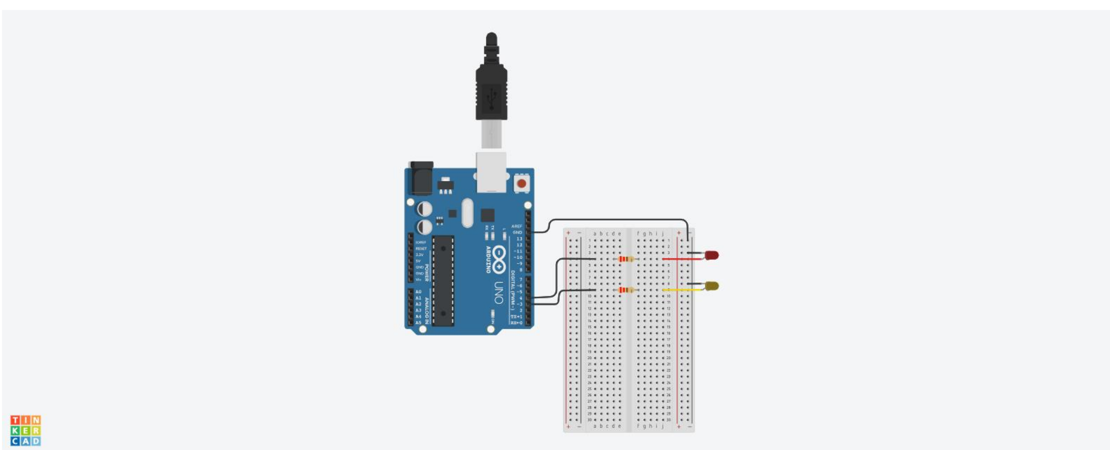

Figure 1: Tinkercad circuit simulation of the Railway Crossing Flasher

7. Working Principle

The system operates on a simple alternating digital logic sequence controlled entirely in software:

- The red LED is switched ON while the yellow LED remains OFF for 500 milliseconds.
- The red LED is then switched OFF while the yellow LED is switched ON for another 500 milliseconds.

- This sequence repeats continuously inside the Arduino's main loop, producing a steady alternating flash — exactly as seen at a real railway crossing signal.

Because only one LED is high at any given moment, the two output pins are effectively driven in a complementary (mutually exclusive) digital pattern, which is the core logic behind the flasher. The 500 ms delay values can be adjusted to change the flash rate without altering the underlying logic.

8. Arduino Source Code

The following C++ (Arduino) code was uploaded to the Arduino Uno to control the LED flashing sequence:

```
// Railway Crossing Flasher Code
const int redLED = 4;    // Red LED connected to Pin 4
const int yellowLED = 3; // Yellow LED connected to Pin 3

void setup() {
  // Set both pins as outputs
  pinMode(redLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
}

void loop() {
  // Red ON, Yellow OFF
  digitalWrite(redLED, HIGH);
  digitalWrite(yellowLED, LOW);
  delay(500); // 0.5 second wait

  // Red OFF, Yellow ON
  digitalWrite(redLED, LOW);
  digitalWrite(yellowLED, HIGH);
  delay(500); // 0.5 second wait
}
```

9. Code Explanation

- redLED and yellowLED store the digital pin numbers (4 and 3) used to identify the two LEDs in code.
- setup() runs once and configures both pins as OUTPUT, so the Arduino can drive them HIGH or LOW.
- loop() runs repeatedly forever, first setting the red LED HIGH and yellow LED LOW, waiting 500 ms, then reversing the states and waiting another 500 ms.
- digitalWrite() sets a pin to logic HIGH (5V, LED on) or LOW (0V, LED off).
- delay() pauses execution for the given number of milliseconds, controlling the flash timing.

10. 3D Enclosure Design

In addition to the electronic circuit, a physical enclosure was designed using Tinkercad's 3D design tool to represent a realistic railway crossing signal post. The design consists of a vertical pole mounted on a circular base, with a cross-buck (X-shaped) railway crossing sign at the top and two circular light housings — red and yellow — positioned on a horizontal crossbar, mirroring the real LED positions in the circuit.

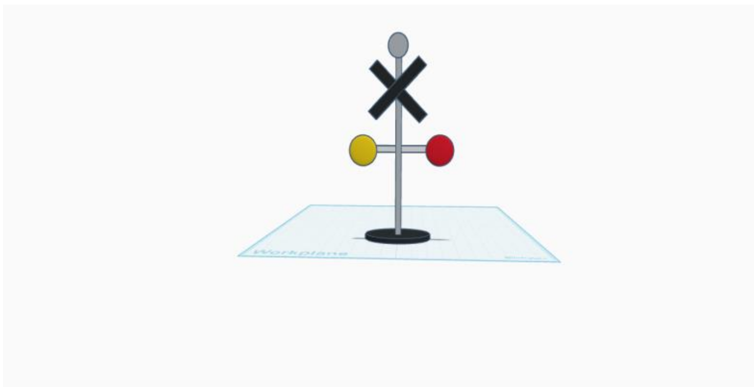

Figure 2: 3D model of the railway crossing signal enclosure (Tinkercad)

The 3D model was exported as an STL file (Railway_Crossing_flasher.stl), making it ready for 3D printing to house the actual Arduino, breadboard, and LED circuit as a complete physical prototype.

11. Applications

- Railway level crossing warning signals for road safety.
- General-purpose alternating hazard/warning light systems (construction zones, barricades).
- Educational demonstration of digital timing logic and microcontroller-based output control.
- Base design for more advanced systems incorporating sensors (e.g., IR/ultrasonic train detection) and automatic gate control.

12. Future Enhancements

- Adding an IR or ultrasonic sensor to automatically detect an approaching train and trigger the flasher.
- Adding a servo-motor controlled crossing gate/barrier that lowers when the flasher is active.
- Adding a buzzer for an audible warning alongside the visual flashing.
- Replacing fixed delay() timing with millis()-based non-blocking timing for smoother, scalable control.

13. Conclusion

The Railway Crossing Flasher project successfully demonstrates how a simple digital logic sequence, implemented through an Arduino microcontroller, can replicate a real-world railway safety signal. By combining circuit simulation, embedded C++ programming, and 3D physical design, the project ties together core Digital Logic Design concepts with a practical, recognizable application. The working simulation confirms that the red and yellow LEDs flash alternately at a fixed 500 ms interval, correctly modeling the visual warning behavior of an actual railway crossing signal.